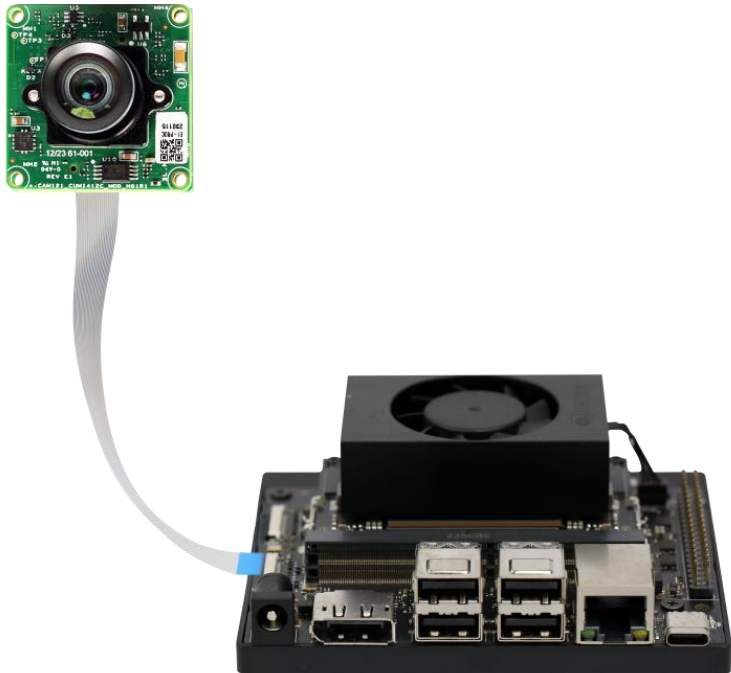


e-CAM121_CUONX

Getting Started Manual



Version 1.2

e-con Systems

8/27/2023



e-con Systems

Think Camera. Think e-con.

Disclaimer

The specifications of e-CAM121_CUONX board and instructions on how to use this board with Jetson Orin Nano™/NX™ development kit are provided as reference only and e-con Systems reserves the right to edit/modify this document without any prior intimation of whatsoever.

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Introduction to e-CAM121_CUONX

e-CAM121_CUONX is a 12 MP camera from e-con Systems, a company with over two decades of experience in designing, developing, and manufacturing OEM cameras. This camera board is directly interfaced with a NVIDIA® Jetson Orin Nano™/NX™ development kit through a CAM1 connector.

e-CAM121_CUONX board consists of a custom lens camera module based on Sony® IMX412-AACK-D CMOS image sensor. This 12 MP color camera has 1/2.3" optical form-factor image sensor with electronic rolling shutter and utilizes Jetson™ platforms in-built image signal processor (ISP). It also comes with an S-mount lens holder (M12 board lens), which is the most commonly used small form-factor lens mount for board cameras and offers customized optics.

e-CAM121_CUONX supported resolutions and frame rates are listed in the following table.

Table 1: Supported Resolutions and Frame Rates of e-CAM121_CUONX in 4-Lane

S.No	Resolution	Frame Rate (fps)
1	4056x3040 (10 BPP)	60
2	4056x3040 (12 BPP)	40
3	4056x3040 (12 BPP)	30
4	4056x2288 (10 BPP)	60
5	4056x2288 (10 BPP)	30
6	2028x1112 (10 BPP)	240
7	2028x1112 (10 BPP)	60

This document describes how to interface the e-CAM121_CUONX board on the Jetson Orin Nano™/NX™ development kit and how to use the e-CAM121_CUONX board.

Prerequisites

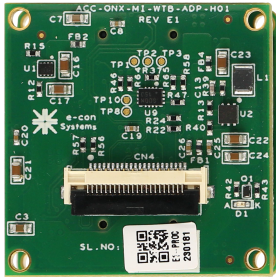
The prerequisites are as follows:

- Host PC with Ubuntu 20.04 (64-bit) or Ubuntu 18.04 (64-bit) to flash the binaries in the Jetson Orin Nano™/NX™ development kit.
- Host PC must contain minimum free space of 60 GB in the directory.

Parts Supplied

The parts supplied with the kit are listed in the following table.

Table 2: Parts Supplied and its Quantity

Parts Supplied	Images	Quantity
Custom Lens Camera Module (e-CAM121_CUMI412C_MOD_H01R1)		1
Adaptor Board (ACC-ONX-MI-WTB-ADP-H01)		1
15cm FPC Cable		1

Description

e-CAM121_CUONX is a multi-board camera, which is compatible with the Jetson Orin Nano™/NX™ development kit. When e-CAM121_CUONX is interfaced with the Jetson Orin Nano™/NX™ development kit, it supports a single camera module with 4-lane MIPI configuration. The module is based on IMX412-AACK-D CMOS image sensor from Sony®. The IMX412 is a 1/2.3" optical form-factor CMOS image sensor with an electronic rolling shutter.

e-CAM121_CUONX is a multi-board solution, which has two boards as follows:

- Camera module (e-CAM121_CUMI412C_MOD_H01R1)
- Adaptor board (ACC-ONX-MI-WTB-ADP-H01)

The front and rear views of the camera module and the adaptor board are shown in the following figures.

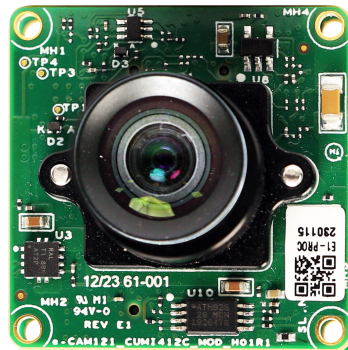


Figure 1: Front View of e-CAM121_CUONX Module Board

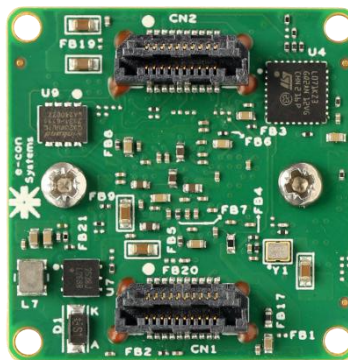


Figure 2: Rear View of e-CAM121_CUONX Module Board



Figure 3: Front View of ACC-ONX-MI-WTB-ADP-H01R1 Adaptor Board

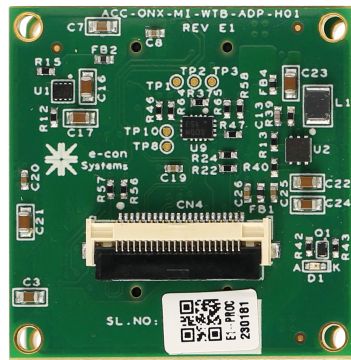


Figure 4: Rear View of ACC-ONX-MI-WTB-ADP-H01R1 Adaptor Board

e-CAM121_CUONX camera module has two 20-pin Samtec connector (CN1 and CN2) for mating with the ACC-ONX-MI-WTB-ADP-H01 adaptor board. This adaptor board acts as a bridge between the camera module and the Jetson Orin Nano™/NX™ development kit. The development kit provides the voltages required for camera module. The adaptor board consists of 22-pin FPC connector (CN4), through which the e-CAM121_CUONX camera module is connected to the Jetson Orin Nano™/NX™ development kit over the FPC cable of maximum 15 cm length.

Board Handling Procedure

This section describes the handling procedure of e-CAM121_CUONX board.

The procedure to assemble camera board to the Jetson Orin Nano™/NX™ development kit is described in the following section.

- [Camera Board Connection](#)
- [Interfacing with Jetson Orin Nano Development Kit](#)
- [Interfacing with Jetson Orin NX Development Kit](#)
- [Pulse Width Modulation Connection](#)

Camera Board Connection

e-CAM121_CUONX is provided with a 15cm FPC cable, for connecting with the Jetson Orin Nano™/NX™ development kit.

The conductive and insulation side location of the FPC cable is shown in the following figure.



Figure 5: FPC Cable Conductive and Insulation Side Location

The steps to connect the camera board are as follows:

1. Unlock the CN4 connector in the adaptor board for inserting the FPC cable.

CN4 connector has a lock actuator, which is used for locking and unlocking the cable as shown in the following figure.



Figure 6: CN4 Connector Location

2. Lift the lock actuator at center with slight force for unlocking the connector as shown in the following figures.



Figure 7: Unlocking CN4 Connector

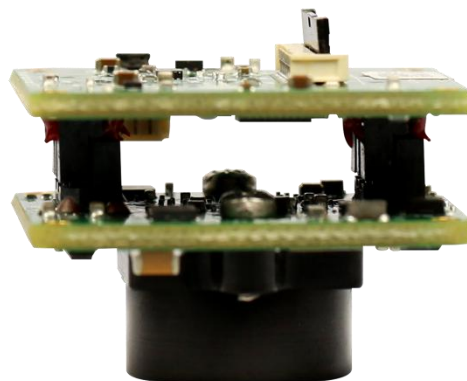


Figure 8: CN4 Unlocked Connector Position

3. Insert the FPC cable to the CN4 connector in the adaptor board.
4. Lock the CN4 connector by pressing the lock actuator from top to bottom.

The conductive side of the cable must face the board as shown in the following figure.



Figure 9: Connector in Locked Position

Note:

- Care must be taken, while connecting the cable to the e-CAM121_CUONX connector.
- If FPC cable is connected in reverse direction to the CN4 connector of e-CAM121_CUONX, it might damage the e-CAM121_CUONX as well as the Jetson Orin Nano™/NX™ development kit.

Interfacing with Jetson Orin Nano Development Kit

Jetson Orin Nano™ development kit has a CAM1 connector to interface with the camera module.

The steps to interface with the camera module are as follows:

1. Insert the FPC cable on the CAM1 connector of the Jetson Orin Nano™ development kit before powering ON the Jetson Orin Nano™ development kit.

The CAM1 connector location in the Jetson Orin Nano™ development kit is shown in the following figure.

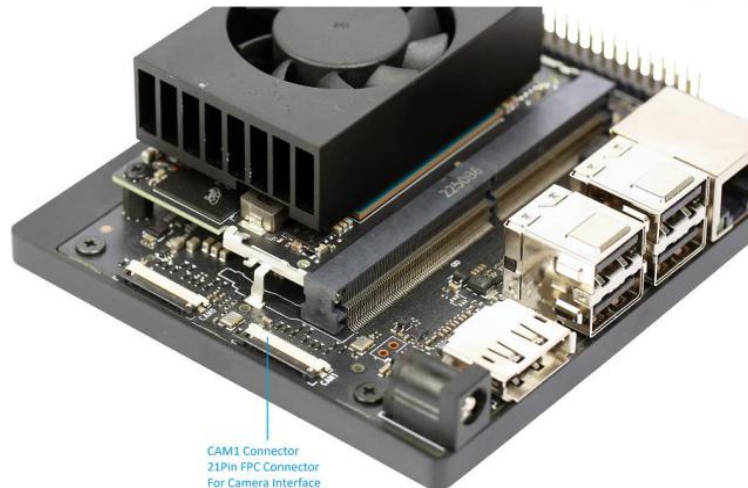


Figure 10: Camera Connector Location in Jetson Orin Nano Development Kit

2. Unlock the CAM1 connector for inserting the FPC cable.

The location of actuator in the Jetson Orin Nano™ development kit is shown in the following figure.

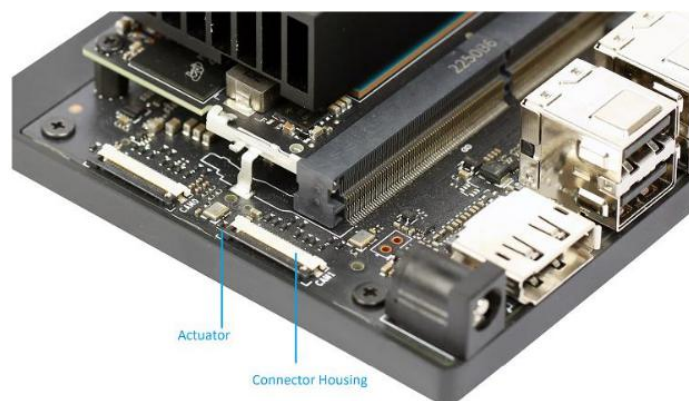


Figure 11: Jetson Orin Nano Kit Camera Connector Actuator Location

3. Pull out the actuator at centre with slight force for unlocking the connector as shown in the following figures.



Figure 12: Unlocking Jetson Orin Nano Development Kit Camera Connector

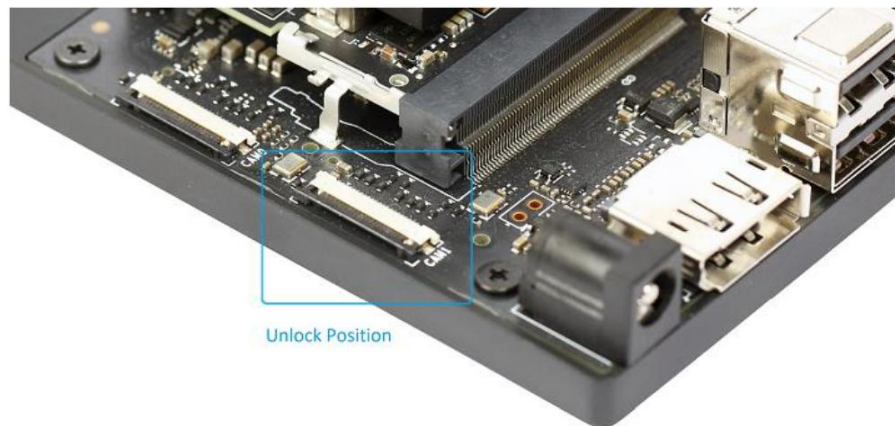


Figure 13: Unlocked Position of Jetson Orin Nano Development Kit Camera Connector

4. Insert the FPC cable to the CAM1 connector of Jetson Orin Nano™ development kit as shown in the following figure.

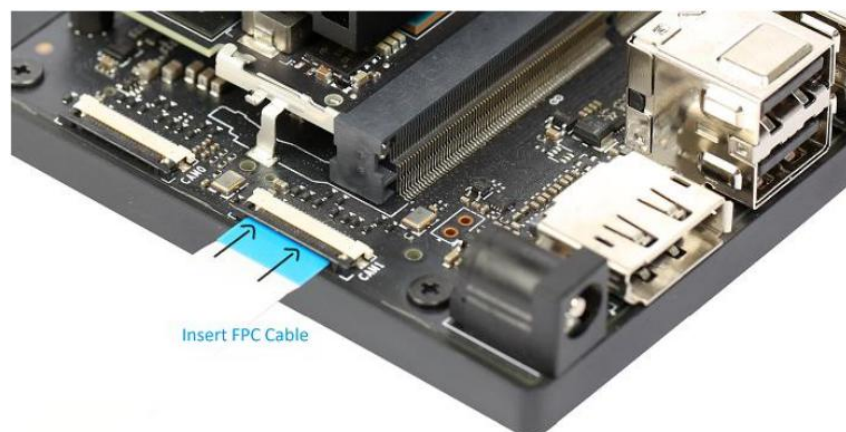


Figure 14: FPC Cable Insertion to CAM1 Connector of Jetson Orin Nano Development Kit

The FPC cable must be inserted in the conductive side of the cable to face the board side as shown in the following figure.

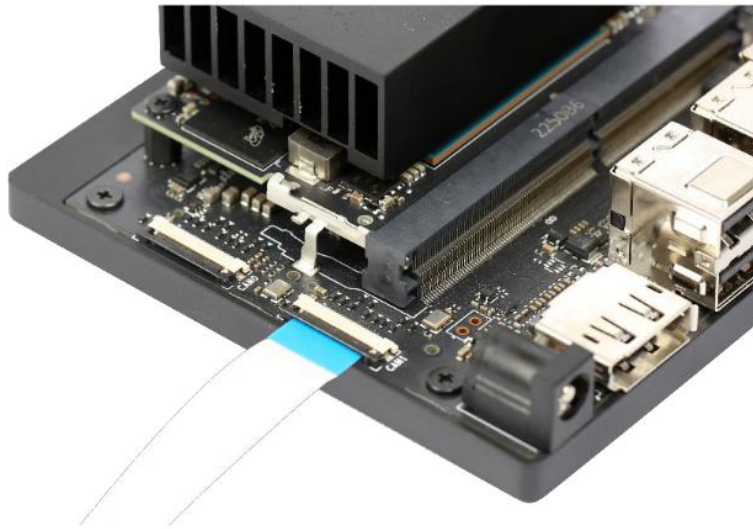


Figure 15: FPC Cable inserted to Jetson Orin Nano Development Kit

5. Lock the CAM1 connector by pressing both ends of actuator with same equal force as shown in the following figures.

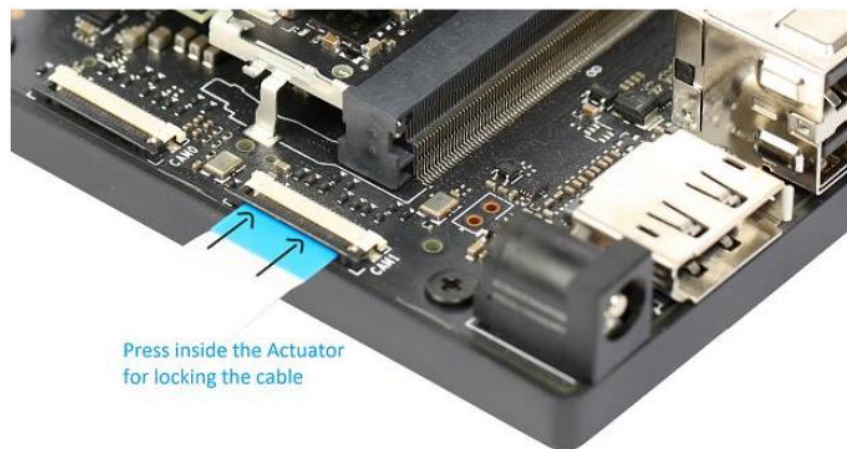


Figure 16: Locking FPC Cable

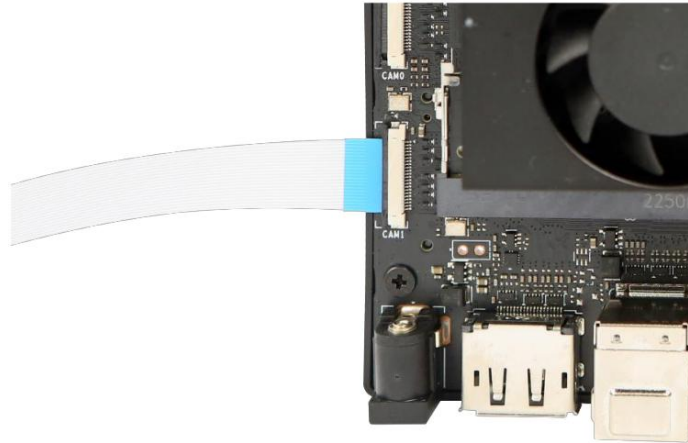


Figure 17: FPC Cable Locked Position on Jetson Orin Nano Development Kit

Now, e-CAM121_CUONX is connected to the Jetson Orin Nano™ development kit as shown in the following figure.

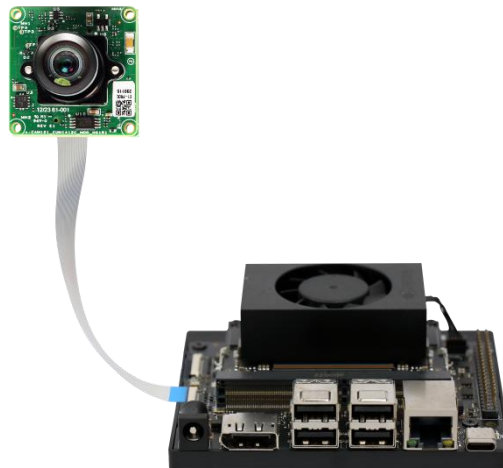


Figure 18: e-CAM121_CUONX Setup Interfaced to Jetson Orin Nano Development Kit

Jetson Orin Nano™ development kit can be powered either from USB Type-C connector or DC power jack connector. Since USB Type-C is used for low power applications, e-CAM121_CUONX supports only low resolutions in this power supply. To utilize the full listed resolutions of the e-CAM121_CUONX camera with the Jetson Orin Nano™ development kit, it is recommended to connect a 19V power supply to the DC jack.

The power supply connector in the Jetson Orin Nano™ development kit is shown in the following figure.



Figure 19: Power Supply Connectors in Orin Nano Development Kit

6. Connect 19V power supply to DC jack to power ON the Jetson Orin Nano™ development kit as shown in the following figure.

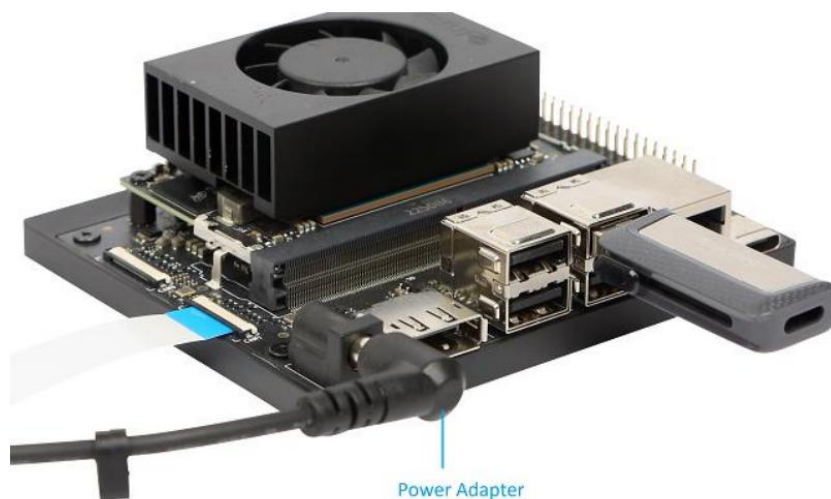


Figure 20: Power Supply Connected to Jetson Orin Nano Development Kit

After powering ON the Jetson Orin Nano™ development kit, the greenish yellow color LED on the Jetson Orin Nano™ development kit will glow. This serves as an indication for power-up of base board as shown in the following figure.

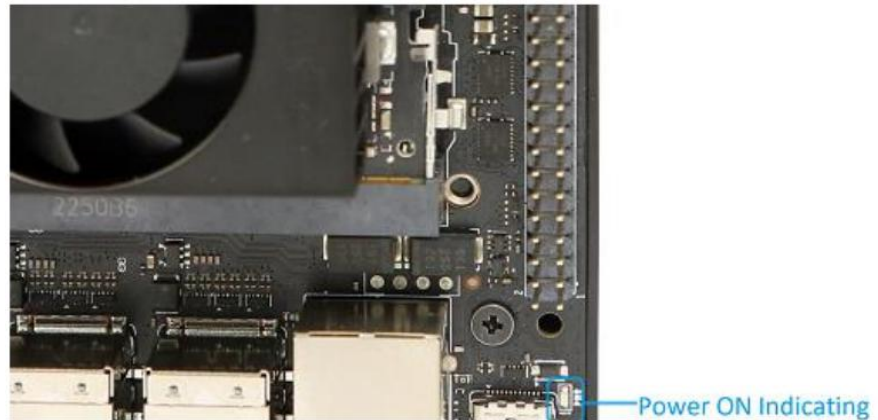


Figure 21: Status LED Indicating Jetson Orin Nano Development Kit Powered ON

Interfacing with Jetson Orin NX Development Kit

The steps to interface the camera module with the Jetson Orin™ NX development kit are as follows:

1. Place the Jetson Orin™ NX SOM on the development kit as shown in the following figure.

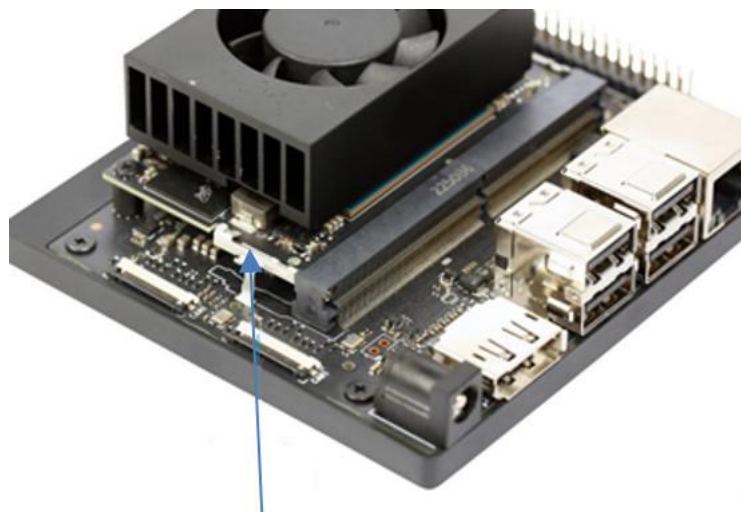


Figure 22: Placing Jetson Orin NX SOM on Development Kit

2. Follow the steps 1 to 6 in *Interfacing with Jetson Orin Nano Development Kit* section for connecting the FPC cable between the Jetson Orin™ NX development kit and the camera module and for powering the Jetson Orin™ NX development kit.

Pulse Width Modulation Connection

Use external trigger signal connector (CN8) for providing PWM trigger signal from external trigger source as shown in below figure.

CN5 position(External Trigger header)

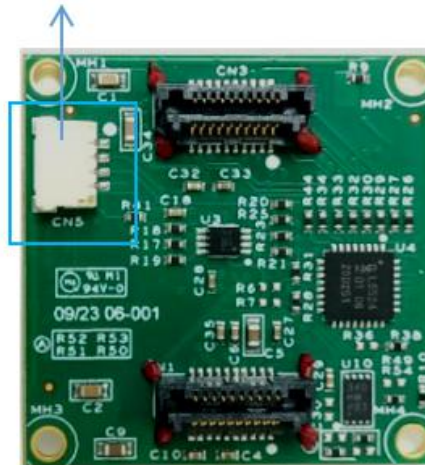


Figure 23: Connector for External Trigger Source.

Pin description is listed in the below table:

CN5 Pin No	Signal Name	Pin Type	Description
1	VCC_3P3	POWER	3.3V Power supply from module
2	EXT_TRIGGER	INPUT	Camera Trigger signal
3	CAM_SHUTTER	OUTPUT	Shutter signal
4	GND	POWER	Ground signal for digital and analog

Note:

Please ensure to provide trigger signal of 3.3V or 5V for proper functionality. The voltage signal greater than 5V will cause permanent damage to the chip.

For detailed information, please refer to the e-CAM121_CUONX_eCAM_Argus_App_User_Manual_Rev_<ver>.pdf.

Software Quick Setup

This section describes the extracting and flashing of e-con Systems provided binaries.

For getting the **e-CAM121_CUONX_JETSON_ONX_ONANO_<L4T>_<DATE>_<VER>.tar.gz** release package, please write to techsupport@e-consystems.com.

The commands and output messages in this manual are represented by different colors as shown in the following table.

Table 3: Notation of Colors

Color	Notation
Blue	Commands running in Host PC
Green	Commands running in Development Kit
Orange	Output message in Development Kit

Before extracting and flashing the e-con provided binaries, flash the Jetpack 5.1.2 in Jetson Orin Nano™/NX™ development kit provided by NVIDIA®, using the SDK Manager.

The steps to setup the Jetson Orin Nano™/NX™ development kit for e-CAM121_CUONX camera are as follows:

1. Download the SDK Manager using the link <https://developer.nvidia.com/nvidia-sdk-manager>.
2. Sign up to an account in NVIDIA® developer site to use the SDK Manager.
3. Run the following command to install the SDK Manager in the host PC.

```
sudo apt-get install ./sdk-manager- [version].[build#].deb
```

After installing the SDK Manager in the host PC, follow the instructions in the <https://docs.nvidia.com/sdk-manager/index.html> link to flash the Jetson Orin Nano™ development kit.

4. Copy the release package into the HOME Directory of the flashed Jetson Orin Nano™/NX™ development kit.

Steps for Extracting and Flashing binaries

The steps to extract and flash the binaries are as follows:

1. Run the following commands to extract the release package in the Jetson Orin Nano™/NX™ development kit to obtain the binaries.

```
tar -xaf e-CAM121_CUONX_JETSON_ONX_ONANO_<L4T_version>_<release_date>_<release_version>.tar.gz  
cd e-CAM121_CUONX_JETSON_ONX_ONANO_<L4T_version>_<release_date>_<release_version>
```

The folder contains the necessary tools to immediately flash the binaries in Jetson Orin Nano™/NX™ development kit with the kernel, camera drivers and applications.

2. Run the following commands in the Jetson Orin Nano™/NX™ development kit to install the e-con provided binaries with support for the e-CAM121_CUONX.

```
sudo chmod +x ./install_binaries.sh  
sudo -E ./install_binaries.sh
```

This script will reboot the Jetson Orin Nano™/NX development kit automatically after installing the binaries successfully.

Steps for Launching Application in Development Kit

The steps to launch the application in the development kit are as follows:

1. Run the following command to confirm whether the camera is initialized.

```
sudo dmesg | grep -i "Detected eimx412 sensor"
```

The output message appears as shown below.

```
Detected eimx412 sensor
```

2. Run the following command to check the presence of camera video node.

```
ls /dev/video*
```

The output message appears as shown below.

```
/dev/video*
```

where (*) represents the number of cameras connected.

The number of times the output message displayed above must be equal to the number of cameras connected to the Jetson Orin Nano™/NX™ development kit.

- Run the following max-clocks shell scripts command before launching the eCAM_argus_camera application in the Jetson Orin Nano™/NX™ development kit.

```
$ sudo ~/max-isp-vi-clks.sh
```

- Run the following command to launch the sample camera application and validate camera.

```
eCAM_argus_camera
```

When the application is launched, you can view the screen similar to the screen shown below.

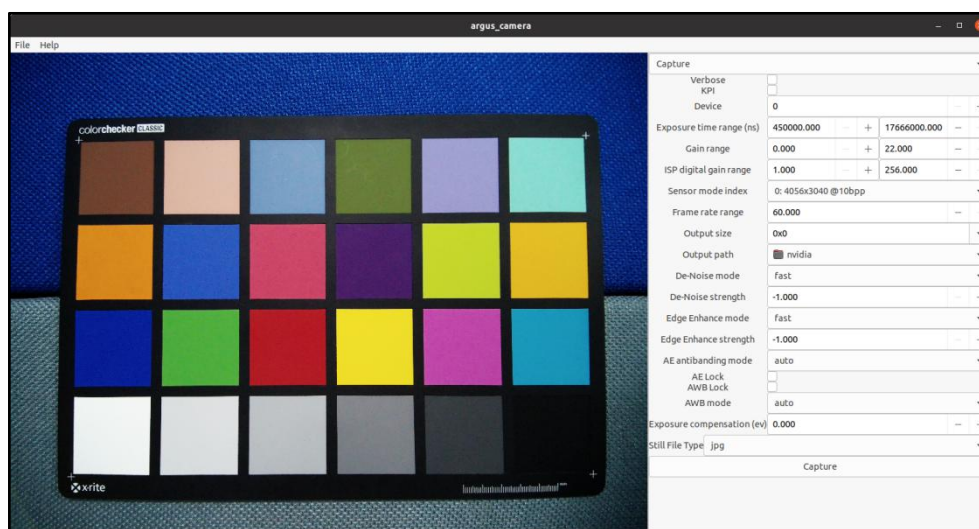


Figure 23: Initial Window when Application is Launched

Note: The default login credentials for Jetson Orin Nano™/NX development kit is listed in the following table.

Table 4: Default Login Credentials

Fields	Inputs
Username	nvidia
Password	nvidia

Reference Documents

This section describes the software and hardware documents of the e-CAM121_CUONX. You can download the software and hardware documents from [Developer Resources](#) website.

Software Documents

The software documents and its description are listed in the following table.

Table 5: Description of Software Documents

S.N O	What I need	Documents to Refer
1	View the camera stream and change the camera controls using GUI.	e-CAM121_CUONX_e-CAM_Argus_App_User_Manual_Rev_<ver>.pdf
2	Build and install the GUI based eCAM_argus_camera application.	e-CAM121_CUONX-eCAM_Argus_Camera_Build_and_Install_Guide_Rev_<ver>.pdf
3	Use GStreamer to control the e-CAM121_CUONX camera on the Jetson Orin Nano™ / NX™ development kit.	e-CAM121_CUONX_GStreamer_Usage_Guide_Rev_<ver>.pdf
4	Build custom kernel with support for using e-CAM121_CUONX.	e-CAM121_CUONX_Developer_Guide_Rev_<ver>.pdf
5	Information about the directory structure and contents of the release package for e-CAM121_CUONX.	e-CAM121_CUONX_Release_Package_Manifest_Rev_<ver>.pdf

Hardware Documents

The hardware documents and its description are listed in the following table.

Table 6: Description of Hardware Documents

S.N O	Documents Name	Description
1	e-CAM121_CUONX_Datasheet_Rev_<ver>.pdf	Describes the feature, connector pin-out details and mechanical dimensions of e-CAM121_CUONX.
2	e-CAM121_CUMI412C_MOD_Datasheet_Rev_<ver>.pdf	Describes the features and specification of e-CAM121_CUMI412C_MOD camera module.
3	e-CAM121_CUONX_Lens_Datasheet_Rev_<ver>.pdf	Describes the optical specification of lenses used in e-CAM121_CUONX.

Known Issues

In this section, you can view the known issues.

Camera streaming gets frozen when switching between modes via the Argus application through the sensor mode index option.

This is a known issue. To successfully stream the camera without encountering freezing issues when switching between modes, follow these steps:

1. Stop the camera streaming.
2. Restart the camera by specifying the desired mode through the command line using the below command line option..

```
eCAM_argus_camera -d <n> --sensormode=<m>
```

Replace **<n>** and **<m>** with the number of video device node and mode number from which you need to stream.

There are three mode options with this release,

mode 0 - 4056x3040 - 10bit @ 60fps

mode 1 - 4056x3040 - 12bit @ 40fps

mode 2 - 4056x3040 - 12bit @ 30fps

mode 3 - 4056x2288 - 10bit @ 60fps

mode 4 - 4056x2288 - 10bit @ 30fps

mode 5 - 2028x1112 - 10bit @ 240fps.

Mode 6 - 2028x1112 - 10bit @ 60fps.

Troubleshooting

In this section, you can view the commonly occurring issues and their troubleshooting steps.

Can I connect the camera to any one of the camera port available in the Jetson Orin Nano™/NX™ development kit?

Currently, support is available for 4-lane connectivity on the Jetson Orin Nano™/NX™ development kit. Therefore, you can only connect the e-CAM121_CUONX camera to the CAM1 connector.

1. Do e-con Systems have any plan to support longer length cable?

e-con Systems provide a very flexible 15 cm FPC cable along with this kit. For customization, please write to camerasolutions@e-consystems.com with your requirement.

2. Does e-CAM121_CUONX camera support OpenCV?

e-CAM121_CUONX works using Video for Linux version 2 (V4L2) APIs and is V4L2 compliant. So, any V4L2 based application can be used to access this camera. OpenCV is also compatible since it uses V4L2 to access the camera.

Please refer to https://www.e-consystems.com/Articles/Camera/accessing_cameras_in_opencv_with_high_performance.asp for detailed information about OpenCV support in e-con Systems cameras.

3. How can I get the updated package?

Please login to the [Developer Resources](#) website and download the latest release package.

What's Next?

After understanding the specifications of camera board and instructions on how to use this board with Jetson Orin Nano™/NX development kit, you can refer to the following documents to understand more about e-CAM121_CUONX.

e-CAM121_CUONX e-CAM Argus App User Manual

e-CAM121_CUONX Developer Guide

API: Application Program Interface.

CMOS: Complementary Metal Oxide Semiconductor.

GUI: Graphical User Interface.

ISP: Image Signal Processor.

L4T: Linux for Tegra.

LED: Light Emitting Diode.

MIPI: Mobile Industry Processor Interface.

USB: Universal Serial Bus.

V4L2: Video for Linux version2 is a collection of device drivers and API for supporting real-time video capture on Linux systems.

Contact Us

If you need any support on e-CAM121_CUONX product, please contact us using the Live Chat option available on our website - <https://www.e-consystems.com/>

Creating a Ticket

If you need to create a ticket for any type of issue, please visit the ticketing page on our website - <https://www.e-consystems.com/create-ticket.asp>

RMA

To know about our Return Material Authorization (RMA) policy, please visit the RMA Policy page on our website - <https://www.e-consystems.com/RMA-Policy.asp>

General Product Warranty Terms

To know about our General Product Warranty Terms, please visit the General Warranty Terms page on our website - <https://www.e-consystems.com/warranty.asp>

Revision History

Rev	Date	Description	Author
1.0	24-May-2023	Initial draft	Product Marketing Team
1.1	01-July-2023	Support for both Orin Nano™ and Orin™ NX development kit.	Product Marketing Team
1.2	27-August-2023	Jetpack updation from 5.1.1 to 5.1.2 and supported Synchronous mode and 4K and 1080p resolution	Product Marketing Team